

Final Report
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Sea Scallop Research Set-Aside Program

17-SCA-24

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Project Title: Factors Influencing Scallop Landings per Unit Effort (LPUE)

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Amount: We were granted 22,517 pounds of scallops (\$270,199) for research and compensation.

Project Summary

The School for Marine Science and Technology (SMAST) at the University of Massachusetts Dartmouth collaborated with members of the US Limited Access scallop fleet to develop a tool for improved landings per unit effort (LPUE) estimates. More accurate LPUE projections will lead to less uncertainty in setting days at sea allocations. The 2017 Scallop Research Set-Aside Program award 17-SCA-24 supported the planning, meeting coordination, and analyses associated with gathering information and model-building.

Project Description

Objectives

The goal of this project is to reduce potential management uncertainty in days-at-sea allocation by improving the estimate of scallop landings per unit effort (LPUE) in open areas. We incorporated fishermen's input into models that account for environmental and behavioral factors.

Methodology

We hosted meetings with fishermen to discuss project objectives and gain insight into factors to include in our predictive models. A combination of in-person meetings and written (email) surveys were used for data collection. We requested and received dealer logbook, VTR, and NEFOP data from GARFO based on factors indicated by fishermen/vessel owners. We used generalized linear models (GLMs) to model the LPUE based on those factors. We compared the modeled LPUE estimates to the observed LPUE and the LPUE predictions used in developing specifications (i.e. days at sea allocations) in recent years.

Timeline

The project began on 1 March 2017, and continued through the 2017 and 2018 fishing years. We requested and were granted a No Cost Extension to complete analysis with feedback from the Scallop Assessment Working Group.

Project Results

We held a meeting in New Bedford, MA with members of the scallop fishing industry (fishermen, vessel owners, and support staff) and distributed surveys (Supplemental Document 1) via email to vessel owners for circulation among captains. Meeting participants suggested that catch rates are most likely to show a relationship with LPUE at low catch rates. Attendees agreed that price, size, yield, shell height-meat weight relationship were major important factors as well as the skill level of the captain and crew. Participants also noted that gear requirements and gear condition can play a role in deciding where to fish or how long to stay out on a trip. To a lesser degree than other factors, some participants agreed weather may impact fishing decisions. The attendees of this meeting were all involved in New Bedford-based operations and agreed that home port makes a difference in fishing style and LPUE. They confirmed that we should seek additional input from industry members from other ports, especially ports to the south. We received 10 responses to our survey from Barnegat Light, NJ captains. Results were broadly similar to the results from New Bedford (price, yield, size, crew experience), though weather and distance seemed to get more emphasis among this group. Additionally, several respondents indicated bottom condition is an important factor.

We met with Dvora Hart at the Northeast Fisheries Science Center (NEFSC) to discuss project progress and design and with Ben Galuardi at Greater Atlantic Regional Fisheries Office (GARFO) to discuss data availability and access. In these meetings, we described our project plans and discussed data needs, the current LPUE estimation method, and modeling approaches.

Based on meetings and survey responses, we prepared a data request to obtain VTR/logbook and dealer data records from GARFO. We received the requested data along with some initial data visualizations provided by Ben Galuardi, GARFO Statistician. After initial data exploration, we began building generalized linear models. The preferred model of LPUE included significant effects for year (peaking in 2011), month (generally peaking in April), region, price of the 10-20 count market category (generally decreasing with price), permit type, state of sale port, and the squared proportion of under 10 count (U10) market category scallops in landings. We presented project progress and preliminary results to the Scallop Stock Assessment Working Group (WG). Methods and preliminary results are fully described in the working paper we submitted to the group as an appendix to the assessment document (Supplemental Document 2).

Our model performed similarly on average to the model currently used to estimate LPUE for days at sea allocations, but there were some problematic years (2011 and 2017). We identified several potential major contributors to the poor predictions in some years, and we expect addressing these factors would considerably improve performance. Our model as presented to the Working Group did not account for the changes over time in seafloor area available to harvest using DAS (e.g. the conversion of part of the Elephant Trunk area from a closed area to open fishing grounds in 2011). We addressed this by scaling our model predictions to expected changes in scallop density rather than biomass. The updated method improves the model performance on average and substantially improves performance in 2011 (Figure 1). Additionally, our projections in each year are based on status quo fishing patterns relative to the previous year. We suspect that replacing these naïve assumptions with more realistic ones based on expert knowledge (i.e. fishermen's and/or PDT member's insights) would improve the accuracy of projections. Finally, we have been unable to obtain a single consistent source of scallop biomass estimates from 2007-2017, despite assistance from scallop plan development team (PDT) members and repeated requests to the lead scallop assessment scientist through our Cooperative Agreement.

After the conclusion of the SAW when FY 2017 landings data became available, we updated our model and projected 2019 LPUE with status quo expectations (Figure 1 and Supplemental Document 3). Our projected 2019 LPUE is 3,296 pounds per day.

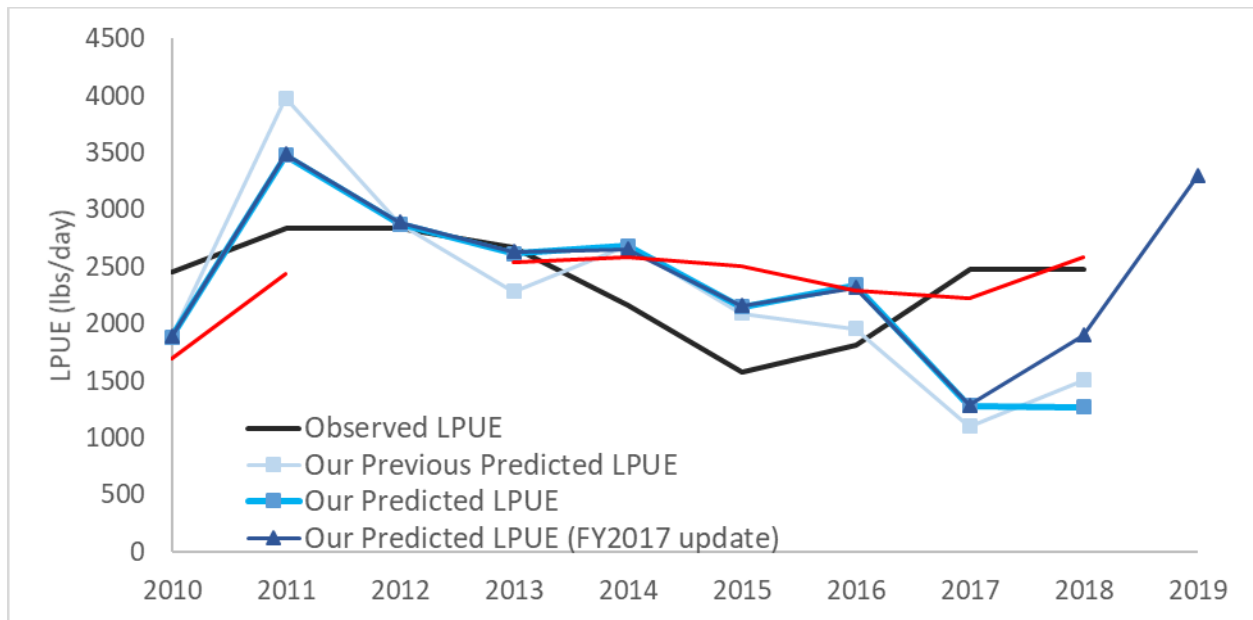


Figure 1. Model predicted LPUE from our original and updated configurations compared to the LPUE estimates historically and currently used in management and to the observed LPUE.

Scallop Assessment Review:

The methods and results of the first phase of modeling were presented at the Scallop Assessment Working Group Data Meeting in Woods Hole, MA in February 2018 (Supplemental Document 4). We solicited feedback on our progress and plans for the rest of the project, and the Working Group suggested several adjustments to our model, including: considering a linear term for the percentage of U10 scallops in landings (we included a square of this value) and adjusting distributional assumption (replacing a log transformation and identity link with raw values and log link in the GLMs). The WG also made suggestions for the evaluation step (retrospectively projecting LPUE): being sure to omit data for the projection year from the data used in model fitting and considering how to address the year effect in the model (scaling the previous year to the perceived change in exploitable biomass).

We incorporated the WG's suggestions and presented our updated results to the Scallop Assessment Working Group in late March 2018 (Supplemental Document 5). The updated model of LPUE was a GLM with Gaussian distributed errors and log link. It included significant effects for year (peaking in 2011), permit type, month (generally peaking in April), region, price of the 10-20 meat count market category (generally decreasing with price), region of sale port, and the squared proportion of U10 count market category scallops in landings (the corresponding linear term was tested, per the WG's suggestion, but it was not retained in the model due to lack of statistical significance). Additional progress at this stage included an out-of-sample validation procedure to support the selected model.

In advance of the third WG meeting in early May 2018, we completed our retrospective projections of LPUE for 2010-2017, using the perceived change in exploitable biomass to scale the year effect, as suggested by the WG. We presented our updated results at the May meeting (Supplemental Document 6). The WG requested a plot of the changes in coefficient values over time, which we

agreed to provide in a finalized version of our working paper. Our preliminary results suggested that our new LPUE estimation method performed similarly to the methods that had previously been used in making days-at-sea allocations. We acknowledged room for improvement by accounting for the change in amount of seafloor available to DAS fishing from year to year (i.e., due to changing Access Areas to Open Area fishing grounds and vice versa), by replacing status quo assumptions of fishing effort distribution with expert judgement, and by replacing a patchwork of exploitable biomass estimates with a single series. Some members of the WG expressed that the method showed promise and expressed support for continuing to improve it.

We updated our working paper to include the requested plots of coefficient values by year and submitted a final draft to the WG chair on 22 May to include as an appendix to the Working Group report (Wright et al., 2018; Supplemental Document 2). The appendix was reviewed by the Stock Assessment Review Committee in June. Based on the WG's recommendations, the model was not included for management in FY 2019 but with subsequent revisions and updated data could be considered for future management.

Program Outreach:

We discussed the project at SMAST Fishermen's Steering Committee meetings in August 2017 and 2018. We attended the Boston Seafood Festival in August and 2018 to present research to the public. We presented project progress and preliminary results to the Scallop Assessment Working Group during meetings in February, March, and May 2018.

In addition to working meetings and public outreach, we presented information related to the project at several scientific meetings (Supplemental Documents 7-9):

Wright BL, CE O'Keefe, B Galuardi, J Peros & SX Cadrin. 2019. Projecting scallop LPUE in the northeast US with fishermen's help. Lowell Wakefield Fisheries Symposium. Anchorage, AK.

Wright BL, CE O'Keefe, B Galuardi, J Peros, & SX Cadrin. 2019. Using fishermen's insights in projections of scallop landings per unit effort (LPUE) in the northeast US. University of Massachusetts Intercampus Marine Science Research Symposium. New Bedford, MA. (poster)

Wright BL, CE O'Keefe & SX Cadrin. 2017. Improving fishery catch rate estimates to reduce management uncertainty in the US sea scallop fishery. ICES CM 2017/S:244.

Expenditures

We allocated the awarded 22,517 pounds to one survey participant and one meeting participant. All compensation fishing was completed by the end of the 2017 RSA compensation fishing period on May 31, 2018. We requested and were granted a No Cost Extension to complete analysis with feedback from the Scallop Assessment Working Group.

References

Wright, B.L., C.E. O'Keefe, B. Galuardi, J. Peros and S.X. Cadrin. 2018. Factors influencing scallop landings per unit effort (LPUE): Appendix A8 In: 65th Northeast Regional SAW Assessment Report. US Department of Commerce, Northeast Fisheries Science Center Reference Document, In Prep.